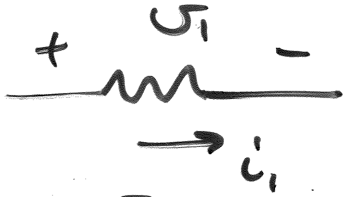
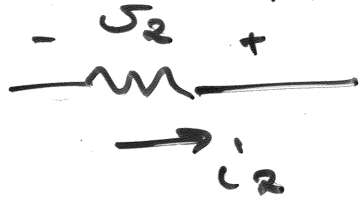
Can you read this?

* <u>Recap:</u>

  → Ohm's Law



$v_1 = +i_1 R$



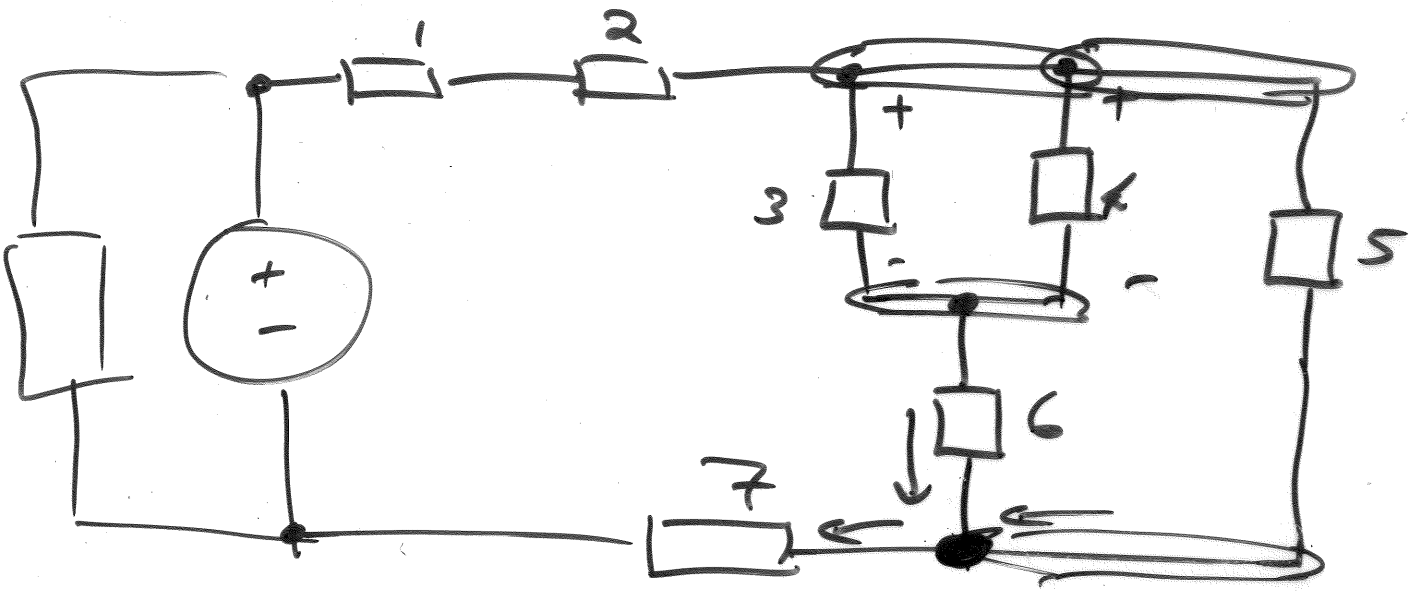
$v_2 = -i_2 R$

* Series: * share same I
  * share single node exclusively.

* ∥ : * share same V
  * share same two nodes at either ends of the branches.

* KCL: (nodes)

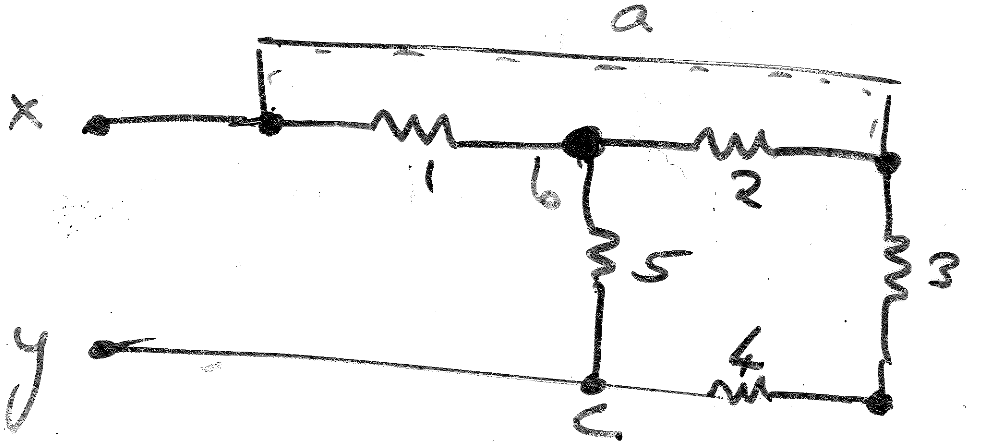
$$\sum I_{IN} = \sum I_{OUT}$$



1 series 2.

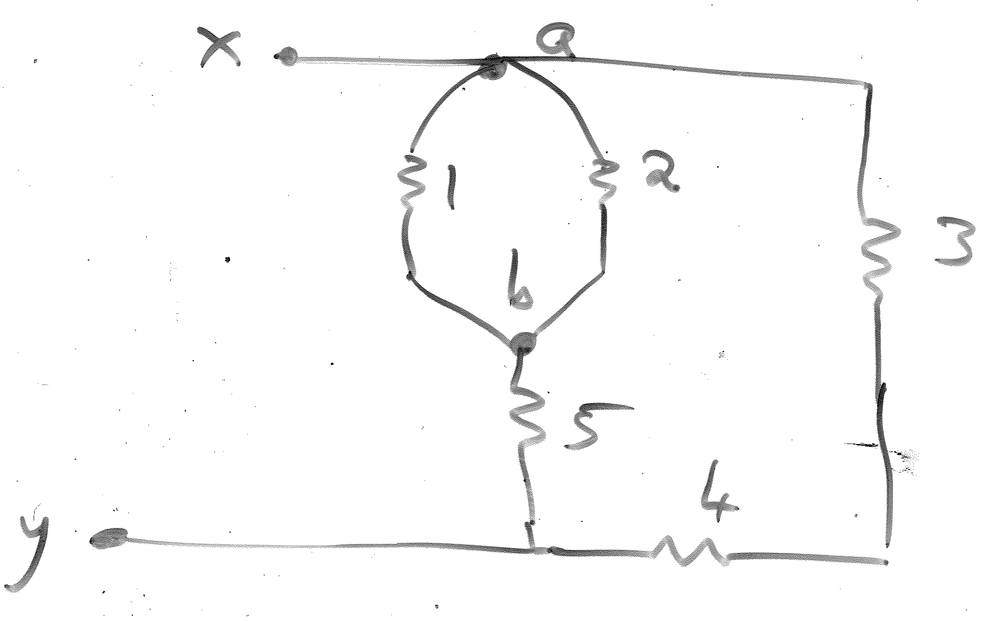
$4 \parallel 5$

$3 \parallel 4$

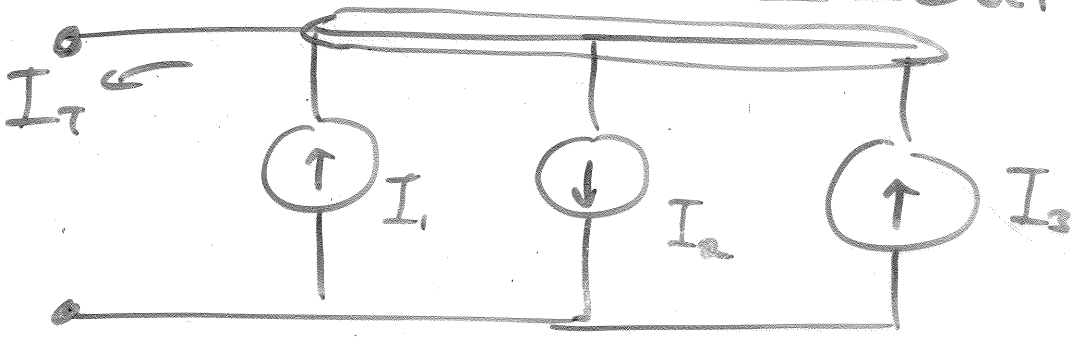


3 s 4

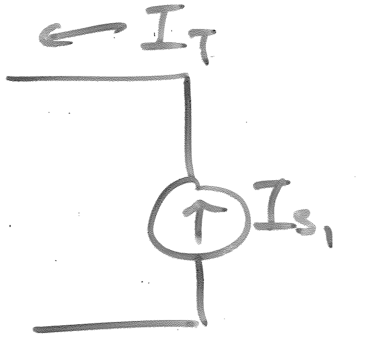
$1 \parallel 2$

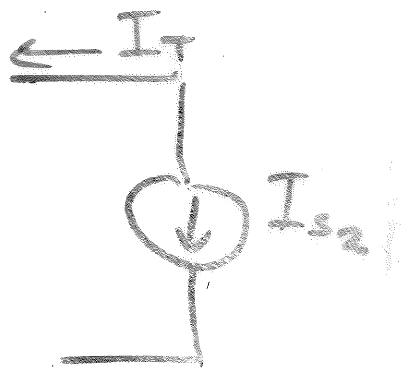


$$\sum I_{IN} = \sum I_{OUT}$$



$$I_1 + I_3 = I_T + I_2$$



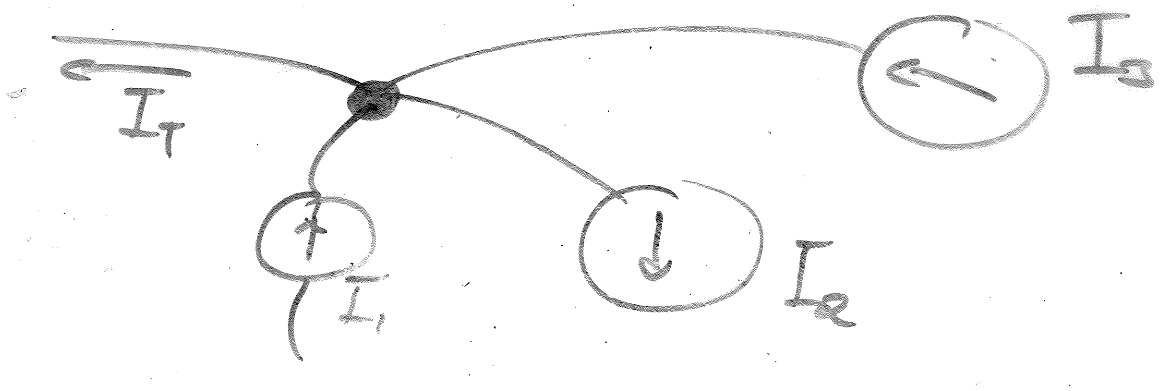


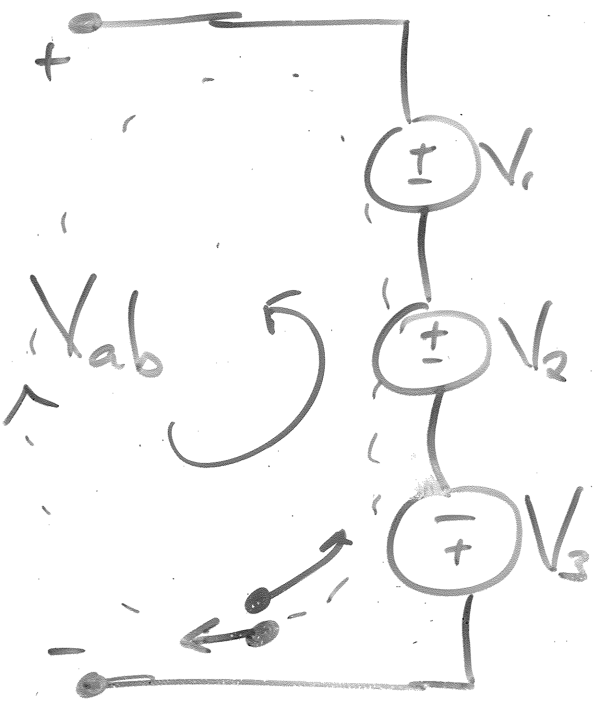
$$I_T = +I_{S_1}$$

$$= I_1 + I_3 - I_2$$

$$I_T = -I_{S_2}.$$

$$= I_2 - I_1 - I_3$$

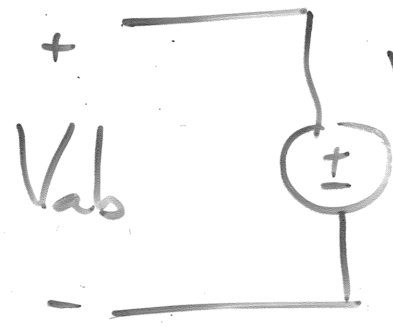




$$\sum V_{LOOP} = 0$$

$$-V_{ab} + V_1 + V_2 - V_3 = 0$$

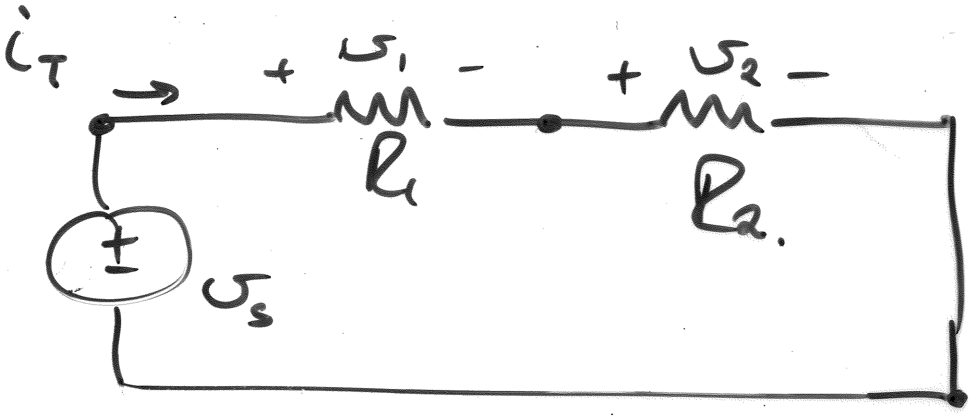
$$+V_3 - V_2 - V_1 + V_{ab} = 0$$



$$V_{s} = +V_{ab}$$

$$= V_1 + V_2 - V_3.$$

* Series R's & Voltage Div



* Ohm: $v_1 = i_T R_1 \qquad v_2 = i_T R_2$

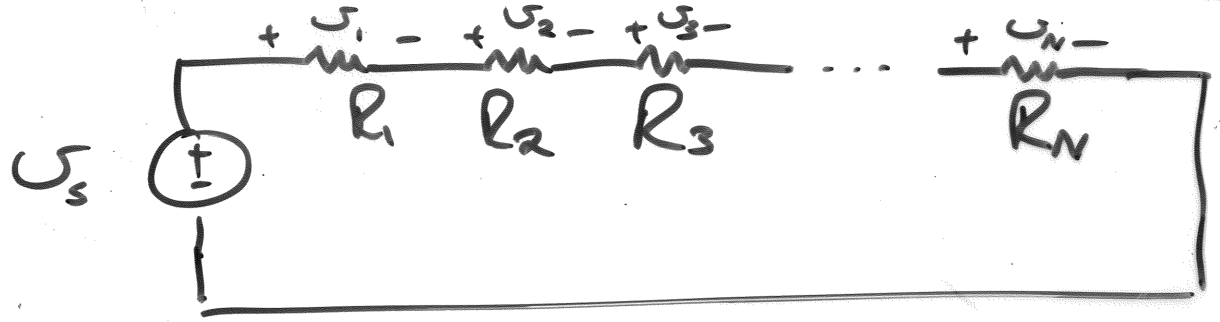
* KVL: $-v_s + v_1 + v_2 = 0.$

$$v_s = i_T (R_1 + R_2)$$

$$\rightarrow \boxed{i_T = \frac{v_s}{R_1 + R_2}}$$

* $$v_1 = \frac{v_s R_1}{R_1 + R_2} \qquad v_2 = \frac{v_s R_2}{R_1 + R_2}$$

$$= \frac{v_s R_1}{R_{eq}} \qquad = \frac{v_s R_2}{R_{eq}}$$



$$R_{eq_{series}} = \sum_{i=1}^{N} R_i \quad \leftarrow \text{Series } R$$

$$v_3 = \frac{v_s \times R_3}{R_{eq}} \quad \leftarrow \text{Voltage Division}$$

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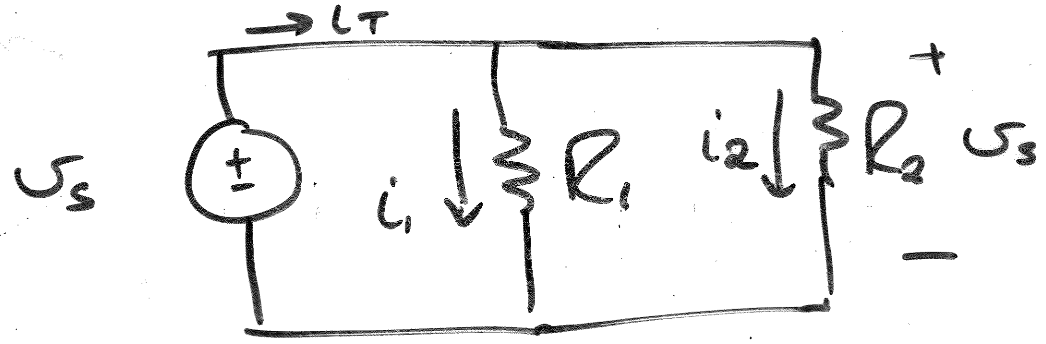
* Parallel R's & Current Division



* Ohm's Law:

$$v_s = \underline{i_1} R_1 \qquad v_s = \underline{i_2} R_2.$$

* KCL: $i_T = i_1 + i_2$.

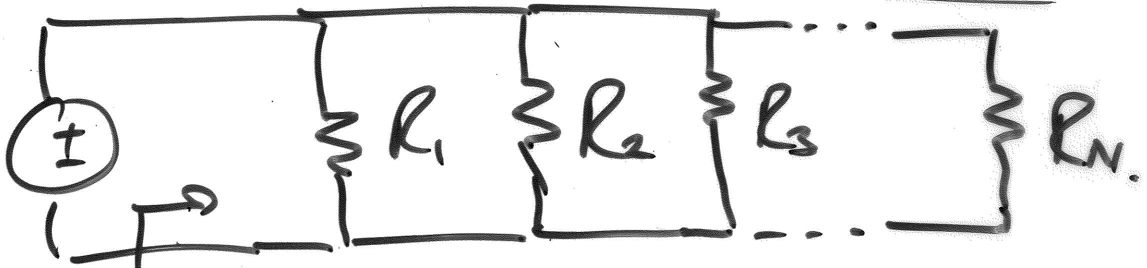
$$i_T = \frac{v_s}{R_1} + \frac{v_s}{R_2} = v_s\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$$\boxed{i = V/R} \implies i_T = v_s / R_{eq}.$$

$$\Rightarrow \quad \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{eq}} = \frac{R_1 + R_2}{R_1 R_2}$$

$R_{\parallel}$ (2 R's) $\rightarrow$ $$\boxed{R_{eq} = \frac{R_1 R_2}{R_1 + R_2}}$$



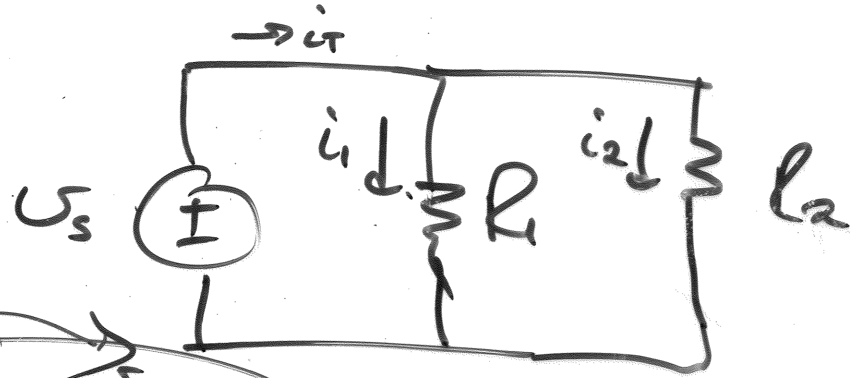
$R_{eq\parallel}$

$$R_{eq} = \cancel{\frac{R_1 \times R_2 \times R_3 \times \ldots R_N}{\Sigma R}}$$

$$\boxed{= \left(\frac{1}{R_1} + \frac{1}{R_2} + \ldots \frac{1}{R_N} \right)^{-1}}$$

$$\boxed{\begin{aligned} +\sum I_{IN} &= +\sum I_{OUT} \\ i_T &= i_1 + i_2 \end{aligned}}$$

Current



$$i_1 = v_s / R_1 \qquad i_2 = v_s / R_2 .$$

$$i_T = i_1 + i_2 = v_s / R_{eq}$$

$$i_T = v_s \times \left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$$R_{eq} = \left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1} = \frac{R_1 R_2}{R_1 + R_2}$$

$$i_T = v_s \times \left(\frac{1}{R_1} + \frac{1}{R_2}\right).$$

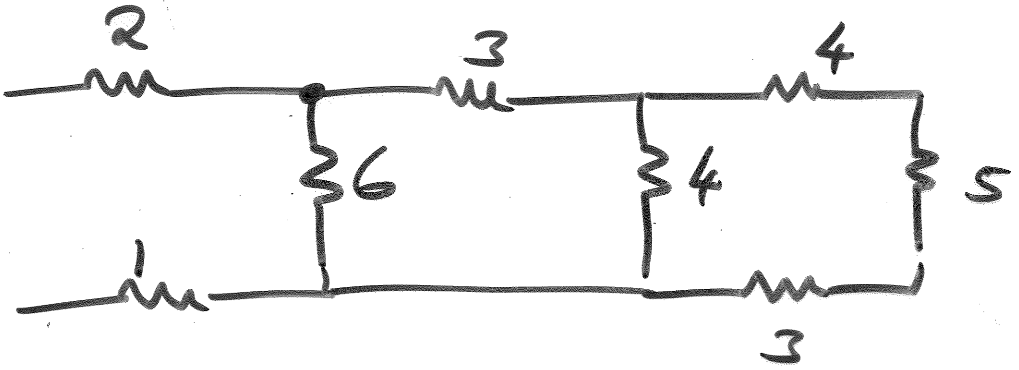
$$v_s = i_T \Big/ \left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$$v_s = \frac{i_T R_1 R_2}{R_1 + R_2.}$$

$$i_1 = v_s / R_1 = \frac{i_T \cancel{R_1} R_2}{R_1 + R_2} \Big/ \cancel{R_1}$$

$$\rightarrow \underline{i_1} = \frac{i_T R_2}{R_1 + R_2.} \quad \Bigg| \quad i_2 = \frac{i_T R_1}{R_1 + R_2.}$$

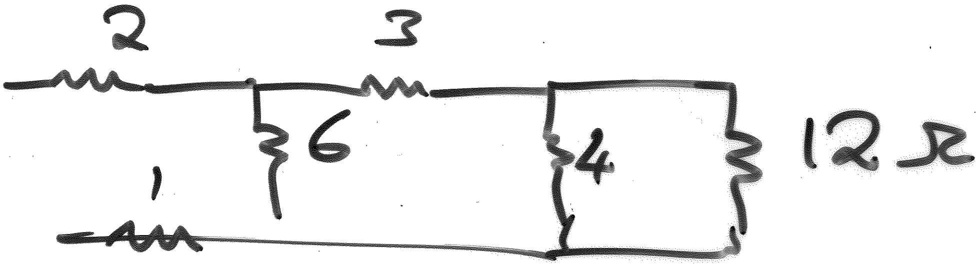
(1)



Ans: $\underrightarrow{6\Omega}$.

$\rightarrow \quad R_{s_1} = 4 + 5 + 3 = 12\Omega.$

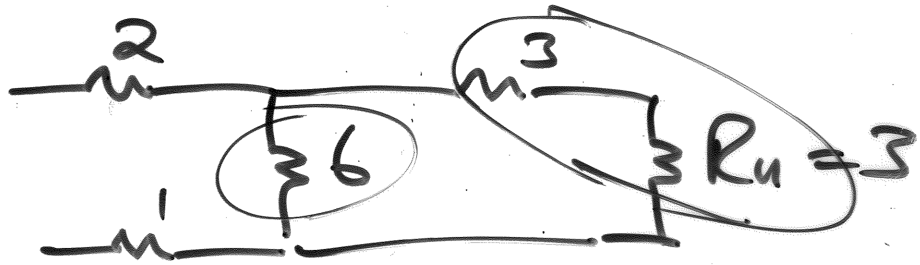
(2)



$$R_{\parallel} = 4 \parallel 12 = \left(\frac{1}{4} + \frac{1}{12}\right)^{-1}$$

$$= \frac{4 \times 12}{4 + 12} = 3\Omega$$

(3)

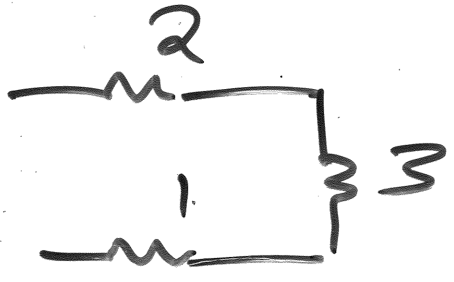


$$R_{s_2} = 3 + 3 = 6\Omega$$

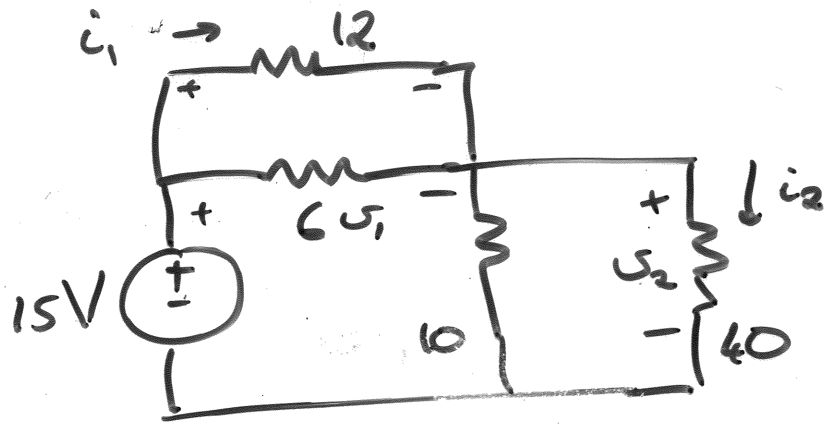
$$R_{\parallel_2} = R_{s_2} \parallel 6 = 6 \parallel 6$$

$$= \frac{6 \times 6}{6 + 6} = 3\Omega.$$

(4)



$$R_{eq} = 1 + 2 + 3 = \underline{6\Omega}$$



$v_1 = 5V$

$v_2 = 10V$

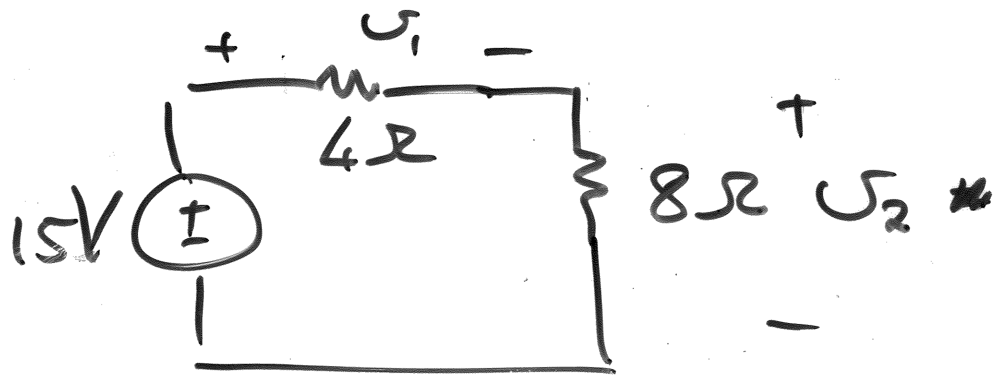
* Ohm: $v_2 = i_2 \times 40$

$v_1 = i_1 \times 12.$

* Series & ||.

$R_{||_1} = 12 // 6 = 4\,\Omega.$

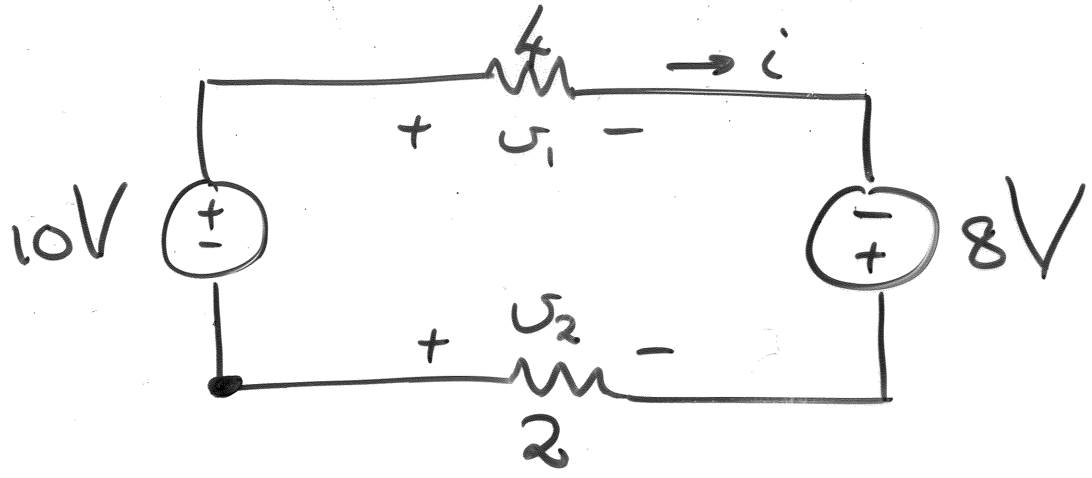
$R_{||_2} = 10 // 40 = 8\,\Omega.$



→ Voltage Div:

$$v_1 = \frac{15 \times R_1}{R_{eq}} = \frac{15 \times 4}{4+8}$$

$$= \underline{5V}$$



* Ohm's Law

$v_1 = \underline{+}4 \times i =$

$v_2 = \underline{-}2 \times i$

* KVL:

$-10 + v_1 - 8 - v_2 = 0$

$$-10 + 4i - 8 - (-2i) = 0$$

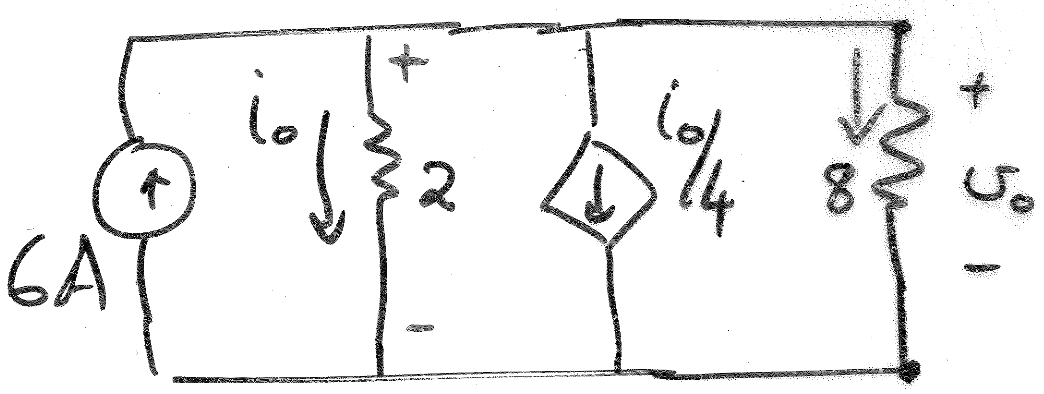
$$-18 + 4i + 2i = 0$$

$$6i = 18$$

$$i = 3A$$

$$v_1 = 4 \times 3 = 12V$$

$$v_2 = -2 \times 3 = -6V$$



$b = 4 \quad n = 2 \quad l = 3$

* Ohm $\quad i_{8\Omega} = +v_o/8$

$v_o = v_{2\Omega} = i_o \times 2.$

* KCL: $\quad \sum I_{IN} = \sum I_{OUT}$

$6 = i_o + \frac{i_o}{4} + i_8$

~~St~~ Subs: $\quad 6 = i_o + i_o/4 + v_o/8$

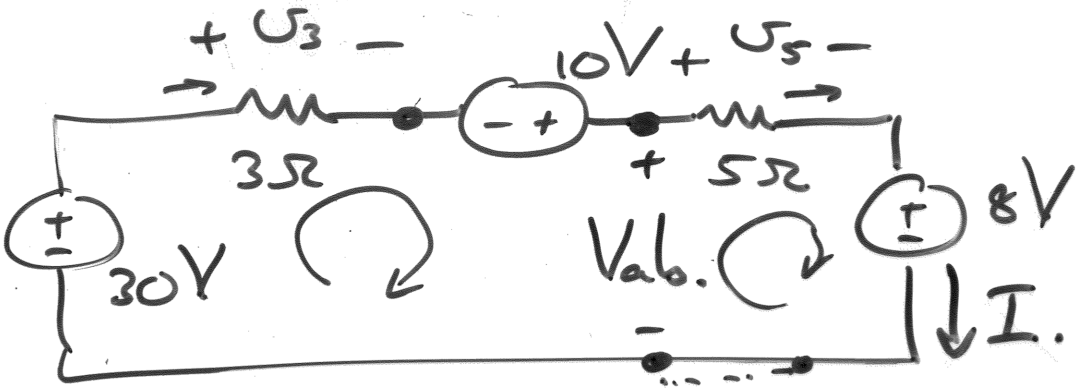
$\longrightarrow \quad 6 = i_o + i_o/4 + 2i_o/8$

$6 = i_o + i_o/4 + i_o/4$

$6 = i_o + \frac{1}{2} i_o$

$6 = \frac{3}{2} i_o$

$i_o = \underline{4A}$



ohm* $\upsilon_3 = +3I$ $\qquad \upsilon_5 = +5I.$

KVL: $-30 + \upsilon_3 - 10 + \upsilon_5 + 8 = 0$

$$-32 + 3I + 5I = 0$$

$$\underline{I = 4A}$$

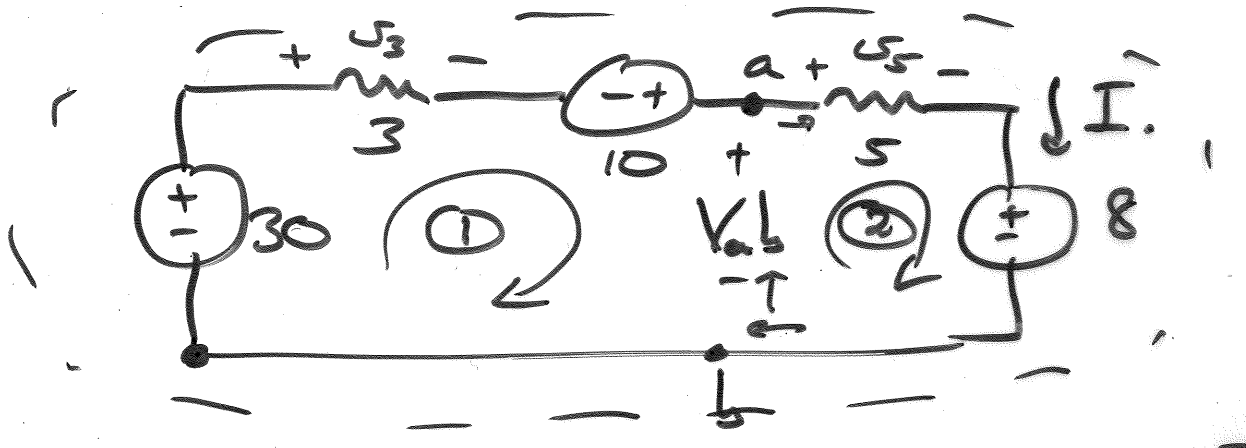
KVL: $-V_{ab} + \upsilon_5 + 8 = 0$

$$-V_{ab} + 5 \times 4 + 8 = 0$$

$$V_{ab} = \underline{28V}$$

* KVL:

$$\Sigma V_{LOOP} = 0$$



$v_3 = +3I$ $\qquad$ $v_5 = +5I.$

① KVL: $-30 + 3I - 10 + V_{ab} = 0$

② KVL: $+5I + 8 - V_{ab} = 0.$

③ KVL: $-30 + 3I - 10 + 5I + 8 = 0.$

$\underline{I = 4A}$

$\underline{V_{ab} = 28V}$